DECK 18 · WEEK 5 · AFTERNOON 2

Options — Payoffs, Strategies & Pricing

From a one-line payoff to the one-period binomial model

CFA Level I · Vol 7 Learning Modules 8–10

Roadmap — Derivatives II: Options — Payoffs, Strategies & Pricing

4 sections · ~2 hours · Built bottom-up from CFA readings

01 **Option payoffs & positions**

Call/put payoffs at expiry · the four basic positions · covered call · protective put · replicating an option

02 **Value drivers & moneyness**

Exercise value + time value · ITM/ATM/OTM from the holder side · no-arbitrage bounds · the six-factor sign table

03 **Put-call parity**

Protective put = fiduciary call · S + p = c + PV(X) · the arbitrage trade · synthetics · forward parity · firm value

04 **One-period binomial**

Up/down gross returns · hedge ratio · risk-neutral probability π · Hightest Capital · where the tree leads next

Learning objectives

By the end of this deck you should be able to:

1. Write the payoff and the profit for all four basic positions — long/short call, long/short put — and say which single one carries an unlimited loss.

2. Split any option premium into exercise value and time value, classify it ITM / ATM / OTM from the holder's side, and check it against the no-arbitrage bounds.

3. Contrast arbitrage and replication for a forward versus an option, and explain why an option's replicating portfolio has to be rebalanced.

4. Name the six factors that move option value and give the sign of each for a call and for a put, including the two that move both the same way.

5. Derive put-call parity from protective put versus fiduciary call, exploit a violation of it, and build any one of the four positions from the other three.

6. Price a European option on a one-period binomial tree by delta hedge and by risk-neutral expectation, and explain why the real-world probability never appears.

SECTION 01

Option payoffs & positions

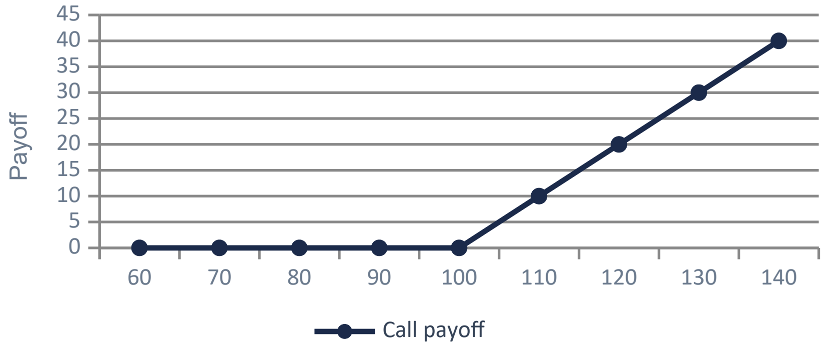
At expiry an option pays Max(0, ST − X) for a call and Max(0, X − ST) for a put — never negative, never linear

The kink at the strike is the whole source of option-market richness: flat on one side of X, one-for-one on the other.

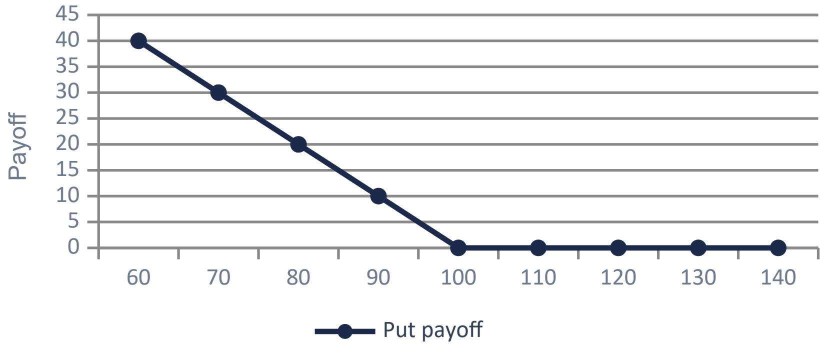
$$cT = \mathrm{Max}(0, ST - X) \qquad pT = \mathrm{Max}(0, X - ST)$$

where: ST = underlying spot at expiry; X = exercise (strike) price. Profit subtracts the premium paid at t = 0: $\Pi = cT - c_0$ (call), $\Pi = pT - p_0$ (put).

Long call payoff — strike 100



Long put payoff — strike 100



Source: CFA Institute, Curriculum Vol 7, Learning Module 8, §6 "Arbitrage", Equations 5–8 and Exhibit 3, printed pp. 180–181.

Four positions, and exactly one of them can lose an unlimited amount: the short call

Buyer and seller are mirror images around the premium: their two profits sum to exactly zero, before transaction costs.

Position	Payoff at T	Max loss	Max gain	Breakeven ST
Long call	Max(0, ST − X)	Premium c_0	Unlimited	$X + c_0$
Short call	−Max(0, ST − X)	Unlimited	Premium c_0	$X + c_0$
Long put	Max(0, X − ST)	Premium p_0	$X - p_0$	$X - p_0$
Short put	−Max(0, X − ST)	$X - p_0$	Premium p_0	$X - p_0$

FOUR INTUITIONS TO LOCK IN

- Buying an option caps your loss at the premium. That cap is what you are paying for.
- Selling an option caps your gain at the premium. You are writing insurance, and you keep the float.
- The short call is the ONLY one of the four with an unbounded loss — a stock has no ceiling. The short put is nasty but bounded: the worst case is a bankruptcy, $X - p_0$.
- The premium never appears in the payoff at T. It is paid at t = 0 and simply shifts the profit line vertically.

[1] Max loss, max gain and breakeven are profit figures; the payoff column is gross of the premium.
[2] A long put hits its $X - p_0$ maximum only if the underlying falls all the way to zero.

Source: CFA Institute, Curriculum Vol 7, Learning Module 8, §6 "Arbitrage", Equations 5–8, printed pp. 181–182.

A covered call sells the upside above X in exchange for the premium — the most-used option overlay in asset management

Long stock + short call. If ST ≤ X you keep the stock and the premium; above X you are called away at X and keep the premium.

$$S_0 - c_0 = X(1+r)^{-T} - p_0 \qquad \text{payoff} = \text{Min}(ST, X)$$

CFA LM 9 EXAMPLE 3 — VFO COVERED CALL ON BIOMIAN

VFO holds non-dividend Biomian at S_0 = INR 295. The CIO sells a six-month call at X = INR 325 for income. r = 4%; the six-month put at the same strike trades at INR 56.

1. Covered call = long stock + short call = $S_0 - c_0$ at inception
2. Parity rearranged: $S_0 - c_0 = X(1 + r)^{-T} - p_0$ (a long bond plus a SHORT put)
3. $295 - c_0 = 325 \times 1.04^{-0.5} - 56 = 318.69 - 56 = 262.69$
4. $c_0 = 295 - 262.69$ = INR 32.31

Sell the call for INR 32.31. Payoff at T caps at X = 325; max profit = 325 − 295 + 32.31 = 62.31.

Source: CFA Institute, Curriculum Vol 7, Learning Module 9, §3 Example 3 "VFO Covered Call Strategy", printed pp. 205–206.

A protective put buys a floor at X for a known upfront cost — the hedge everyone wants and few want to pay for

Long stock + long put. Above X you let the put lapse; below X you exercise and sell at X. A floor at X, paid for with p_0.

$$S_0 + p_0 \quad \text{(protective put)} \qquad \text{payoff} = \text{Max}(ST, X)$$

CFA LM 9 EXAMPLE 1 — PRICING VFO'S SIX-MONTH BIOMIAN PUT AT X = 265

Biomian at S_0 = INR 295, no dividends. A traded six-month call at the same X = 265 is quoted at c_0 = INR 59. r = 4%, T = 0.5.

1. Put-call parity: $S_0 + p_0 = c_0 + X(1 + r)^{-T}$
2. PV(X) = $265 \times 1.04^{-0.5}$ = INR 259.85
3. $295 + p_0 = 59 + 259.85 \Rightarrow p_0 = 318.85 - 295$ = INR 23.85
4. Cross-check both sides: 295 + 23.85 = 318.85 and 59 + 259.85 = 318.85 ✓

Floor at 265 costs 23.85. Max loss = 295 – 265 + 23.85 = 53.85; upside above 265 untouched.

Source: CFA Institute, Curriculum Vol 7, Learning Module 9, §2 Exhibits 1–3 and Example 1 "Biomian Put–Call Parity", printed pp. 200–202.

Replicating an option means borrowing to buy only a FRACTION of the underlying — and rebalancing as moneyness moves

A forward is replicated once and left alone. An option has to be re-hedged. That one difference is why option desks exist.

Position to replicate	Trade at inception (t = 0)	Settlement at t = T if exercised	Rebalance?
Long forward	Borrow PV(F_0); buy the asset	Sell at ST, repay F_0(T)	No
Long call	Borrow part of PV(X); buy hS	Sell at ST, repay X	Yes
Long put	Short hS; lend the proceeds	Receive X, buy back at ST	Yes

WHY ONLY A FRACTION

If exercise were certain you would borrow the full PV(X), exactly as for a forward. It is not — so you borrow only the proportion that matches the likelihood of exercise. That proportion is h, and it is the same h you meet in the binomial model.

WHY h KEEPS MOVING

A forward settles with certainty, so its replicating trades never change. An option's likelihood of exercise moves with every tick, so h moves too. Re-hedging cost is the real economic content of the premium.

AND IN REVERSE

To replicate SELLING a call, short the underlying and lend the proceeds — the exact mirror of buying one. LM 8 practice problem 6 tests precisely this, and the sign is the only thing candidates get wrong.

Source: CFA Institute, Curriculum Vol 7, Learning Module 8, §7 "Replication", Exhibits 4–5, printed pp. 182–184.

SECTION 02

Value drivers & moneyness

An option price splits exactly into exercise value plus time value — and exercise value discounts X, it does not use X

Discounting X rather than using X is the single most common mark-to-market slip in practice — and the standard exam distractor.

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ct = Max(0, St − PV(X)) + Time Value
pt = Max(0, PV(X) − St) + Time Value
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where: PV(X) = X(1 + r)^−(T−t). Time value is always positive and decays to zero at t = T. An out-of-the-money option has zero exercise value, so its price is time value in full.

CFA LM 8 SELF-ASSESSMENT Q3 — GBP EUROPEAN CALL DECOMPOSITION

A three-month European call on a no-dividend stock. X = GBP 50, r = 2%, St = GBP 57.50. The call is quoted at ct = GBP 10.

1. PV of the strike: $X(1 + r)^{-0.25} = 50 \times 1.02^{-0.25}$ = GBP 49.75
2. Exercise value = Max(0, St − PV(X)) = Max(0, 57.50 − 49.75) = GBP 7.75
3. Time value = ct − exercise value = 10.00 − 7.75 = GBP 2.25

7.75 + 2.25 = the GBP 10 premium. Use X instead of PV(X) and you get 7.50 — the trap answer.

Source: CFA Institute, Curriculum Vol 7, Learning Module 8, §3 "Option Exercise Value" and §5 "Option Time Value", Equations 1–4 and Exhibit 2, printed pp. 175–179; Self-Assessment Q3, printed p. 173.

Moneyness is read from the holder's side: a call is in the money when S > X, a put when S < X

Always the buyer's perspective. A short call is never "in the money" — its counterparty's call is, and that is what the term describes.

Moneyness	Call	Put	Sensitivity to S	Typical use
In the money (ITM)	St > X	St < X	Near one-for-one	Stock substitute
At the money (ATM)	St = X	St = X	Moderate	Peak time value
Out of the money (OTM)	St < X	St > X	Very small	Cheap tail hedge

DEEP ITM BEHAVES LIKE THE STOCK

A deep-in-the-money option is highly likely to be exercised, so the curriculum describes "a nearly one-to-one correspondence between option price and underlying price changes".

That is why an ITM call is a financing-efficient way to hold equity exposure: near-identical participation, a fraction of the cash outlay.

DEEP OTM IS ALL TIME VALUE

A deep-out-of-the-money option is very unlikely to be exercised and shows far less price sensitivity for the same move in the underlying. Its exercise value is zero, so what you are buying is entirely time value.

Cheap per contract, and most of it expires worthless. That is not a flaw — it is the price of a tail hedge.

Source: CFA Institute, Curriculum Vol 7, Learning Module 8, §4 "Option Moneyness", Exhibit 1 and Example 2, printed pp. 176–177.

No-arbitrage bounds pin every option: Max(0, S − PV(X)) < c ≤ S, and Max(0, PV(X) − S) < p ≤ X

Price a call above S and I buy the stock and sell the call for free money. Price it below its exercise value and I do the reverse.

$$\mathrm{Max}(0,\ S_t - PV(X)) < c_t \le S_t$$
$$\mathrm{Max}(0,\ PV(X) - S_t) < p_t \le X$$

where: Upper bound: a call buyer will not pay more for the right to buy than the asset itself; a put can never be worth more than the strike it delivers.

CFA LM 8 EXAMPLE 4 — CALL BOUNDS SIX MONTHS INTO A ONE-YEAR CONTRACT

A one-year European call, X = EUR 1,000. At t = 0 the asset trades at S_0 = EUR 990 and r = 1%. Six months later (T − t = 0.5) the spot has risen to St = EUR 1,050.

1. PV(X) over the remaining half-year: $1{,}000 \times 1.01^{-0.5}$ = EUR 995.04
2. Lower bound = Max(0, St − PV(X)) = Max(0, 1,050 − 995.04) = EUR 54.96
3. Upper bound = St = EUR 1,050

54.96 < ct ≤ 1,050. Quote below 54.96 and the call trades under its own exercise value.

Source: CFA Institute, Curriculum Vol 7, Learning Module 8, §6 "Arbitrage", Equations 9–14 and Example 4, printed pp. 181–182.

Six factors move option value; volatility and time move calls and puts the same way, the other four move them apart

Memorise this table once. Level I tests a sign-flip from it every sitting — usually the risk-free rate or the income row.

Factor (an increase in …)	Call	Put	One-line intuition
Value of the underlying (S)	+	–	Calls gain, puts lose as S rises
Exercise price (X)	–	+	Low X is cheap to buy at
Time to expiration (T)	+	+/–	Wider outcome range; put caveat[1]
Risk-free interest rate (r)	+	–	Higher r ⇒ lower PV(X)
Volatility of the underlying	+	+	One-sided payoff likes fat tails
Income / cost of owning S	–/+	+/–	Income goes to the holder, not you[2]

THE PAIRING THE EXAM TESTS

Two factors move calls and puts in the SAME direction: volatility (always) and time to expiration (usually — the exception is a deep-ITM put at a high rate). The other four — underlying, exercise price, risk-free rate and income/carry — move them in OPPOSITE directions.

[1] Longer expiry can LOWER a deep-in-the-money put when the risk-free rate is high (LM 8, printed p. 188).
[2] Income lowers a call and raises a put; a carry cost such as storage or insurance does the reverse.

Source: CFA Institute, Curriculum Vol 7, Learning Module 8, §8 "Factors Affecting Option Value", Exhibits 6–9, printed pp. 186–189.

SECTION 03

Put-call parity

A protective put and a fiduciary call pay the same in every state — so no-arbitrage forces $S_0 + p_0 = c_0 + PV(X)$

Two portfolios that pay the same thing in every state of the world must cost the same today. That is the entire proof.

$$\text{Protective put: } S_0 + p_0 \equiv \text{ Fiduciary call: } c_0 + X(1 + r)^{-T}$$

where: European options only, same X, same T, no income on the underlying, one risk-free rate r. A fiduciary call is a purchased call plus a bond paying X at T.

Portfolio value at t = T	ST < X	ST = X	ST > X
Protective put: $S_0 + p_0$	ST + (X − ST) = X	ST = X	ST + 0 = ST
Fiduciary call: $c_0 + PV(X)$	0 + X = X	X = ST	ST − X + X = ST
Difference	zero	zero	zero
No-arbitrage ⇒ equal cost now	**S + p = c + PV(X)**	**(parity)**	✓

Same payoff in every state ⇒same price today. No distribution, no probabilities, no model — this result is pure arbitrage.

Source: CFA Institute, Curriculum Vol 7, Learning Module 9, §2 "Put–Call Parity", Exhibits 1–4, printed pp. 200–202.

CFA LM 9 Example 1 prices a put straight off a traded call — parity is a pricing engine, not just an identity

Whenever a liquid call and an illiquid put share a name, a strike and a date, parity gives the put its mark — instantly.

$$p_0 = c_0 + X(1 + r)^{-T} - S_0$$

BIOMIAN — A SIX-MONTH PUT FROM A TRADED CALL

Biomian shares at S_0 = INR 295, no dividends. An observed six-month European call at X = INR 265 trades at c_0 = INR 59. r = 4%, T = 0.5.

1. PV of the strike: $X(1 + r)^{-T} = 265 \times 1.04^{-0.5} = 265 / 1.0198$ = INR 259.85
2. Rearrange parity to isolate the put: $p_0 = c_0 + PV(X) - S_0$
3. $p_0 = 59 + 259.85 - 295 = 318.85 - 295$ = INR 23.85
4. Cross-check both sides: $S_0 + p_0 = 318.85$ and $c_0 + PV(X) = 318.85$ ✓

No-arbitrage six-month put premium = INR 23.85, computed without any model of Biomian.

Source: CFA Institute, Curriculum Vol 7, Learning Module 9, §2 Example 1 "Biomian Put–Call Parity", printed p. 202.

CFA LM 9 Example 2 locks in a riskless INR 6.15 a share when the traded put is richer than parity allows

$S_0 + p_0 = 325$ but $c_0 + PV(X) = 318.85$. Sell the expensive portfolio, buy the cheap one, and the difference is yours today.

Leg at t = 0	Cash at t = 0	ST < X	ST = X	ST > X
Sell the underlying at S_0	+295.00	−ST	−ST	−ST
Sell the put at its rich price	+30.00	ST − X	zero	zero
Buy the call	−59.00	zero	zero	ST − X
Buy a bond that pays X at T	−259.85	+X	+X	+X
NET	**+6.15**	**0**	**0**	**0**

READING THE GRID

At t = 0 the investor banks INR 6.15 in cash. At t = T, in every single ending price ST, the four legs net to exactly zero — no residual exposure of any kind, in any state.

Positive cash today, zero risk tomorrow: that is the textbook definition of a riskless arbitrage. Note the profit is exactly the mispricing, 30.00 − 23.85 = 6.15, and it is banked immediately, not at expiry.

Source: CFA Institute, Curriculum Vol 7, Learning Module 9, §2 Example 2 "VFO Put–Call Parity Arbitrage Opportunity", printed p. 203.

Parity rearranges into a synthetic cookbook — each of the four positions is replicable from the other three

Read a row: to build the position on the left, take the combination on the right. Every row is one rearrangement of $S + p = c + PV(X)$.

Position to build	Underlying	Risk-free bond	Call	Put
Underlying (S_0)	–	Long	Long	Short
Risk-free bond (PV(X))	Long	–	Short	Long
Call option (c_0)	Long	Short	–	Long
Put option (p_0)	Short	Long	Long	–

HOW TO REBUILD THIS TABLE INSTEAD OF MEMORISING IT

Start from $S_0 + p_0 = c_0 + PV(X)$ and move one term at a time.

- Solve for S_0: $S_0 = c_0 - p_0 + PV(X)$ → long call, short put, long bond. Row 1.
- Solve for PV(X): $PV(X) = S_0 - c_0 + p_0$ → long underlying, short call, long put. Row 2.
- Solve for c_0: $c_0 = S_0 + p_0 - PV(X)$ → long underlying, long put, short bond. Row 3.
- Solve for p_0: $p_0 = c_0 + PV(X) - S_0$ → long call, long bond, short underlying. Row 4.

Source: CFA Institute, Curriculum Vol 7, Learning Module 9, §3 Exhibit 6 "Replication of Individual Positions under Put–Call Parity", printed p. 205.

A synthetic long stock is long call − short put + PV(X) in cash — same payoff, same cost, no share ever bought

Useful when the cash share is awkward to hold: expensive to borrow, capital-heavy, or restricted. The replicating portfolio bypasses the register.

$$S_0 = c_0 - p_0 + X(1 + r)^{-T}$$

SYNTHETIC LONG STOCK — THE BIOMIAN NUMBERS

The same trade as before: six-month options at X = INR 265, c_0 = 59, p_0 = 23.85, r = 4%, S_0 = INR 295.

1. Buy the call: −INR 59.00
2. Sell the put: +INR 23.85
3. Buy a zero-coupon bond paying X = 265 at T: −INR 259.85
4. Net outlay = 59 − 23.85 + 259.85 = INR 295.00 — exactly S_0
5. At T either branch nets ST: (ST − 265) or −(265 − ST), plus the bond's 265

Costs S_0 = 295 today, pays ST at expiry — indistinguishable from the cash share.

Source: CFA Institute, Curriculum Vol 7, Learning Module 9, §3 Equation 2 and Exhibit 6, printed pp. 204–205.

Put-call forward parity swaps S_0 for the PV of the forward — the version you use when the cash asset is hard to hold

For commodities, currencies and shares with awkward carry, traders price options off the forward rather than the spot.

$$F_0(T)(1+r)^{-T} + p_0 = c_0 + X(1+r)^{-T}$$
$$\Rightarrow \quad p_0 - c_0 = [X - F_0(T)](1+r)^{-T}$$

where: A forward purchase plus a bond of face $F_0(T)$ replicates the underlying. When $F_0(T) = X$ the right side is zero, so $p_0 = c_0$ — the at-the-money-forward equivalence.

CFA LM 9 EXAMPLE 4 — VFO PUT-CALL FORWARD PARITY

The same Biomian trade priced off the forward: $S_0 = 295$, $r = 4\%$, $T = 0.5$, so $F_0(T) = 295 \times 1.04^{0.5}$ = INR 300.84. Still $X = 265$ and c_0 = INR 59.

1. Rearranged forward parity: $p_0 - c_0 = [X - F_0(T)](1 + r)^{-T}$
2. $p_0 - 59 = (265 - 300.84) \times 1.04^{-0.5} = -35.84 / 1.0198 = -35.14$
3. $p_0 = 59 - 35.14$ = INR 23.86

INR 23.86 — the same put, to a rounding penny. Spot parity and forward parity are one equation.

Source: CFA Institute, Curriculum Vol 7, Learning Module 9, §5 "Put–Call Forward Parity", Equation 3, Exhibits 7–8 and Example 4, printed pp. 207–209.

Parity prices a levered firm: equity is a call on the assets, risky debt is a risk-free bond plus a sold put

$V_0 + p_0 = c_0 + PV(D)$, so $V_0 = c_0 + PV(D) - p_0$. Substitute firm value for the underlying and debt face for the strike — parity does the rest.

EQUITYHOLDERS

Payoff at T: Max(0, VT − D)

= long firm assets + a LONG put struck at D
= a LONG CALL on firm value, strike D

- Unlimited upside — the residual claim
- Downside stops at zero: limited liability IS the put
- Both new projects and more volatility raise the call, which is the risk-shifting incentive in one line
- The put shareholders hold is written by the lenders. That single transfer is what leverage really is
- Raise leverage and the call moves closer to the money — good for equity, paid for by the other panel

DEBTHOLDERS

Payoff at T: Min(VT, D) = D − Max(0, D − VT)

= long risk-free bond of face D + a SHORT put

- Upside capped at D — you can only be repaid
- Downside below D on default: that is the sold put
- The credit spread IS the premium on that put, and it widens with the volatility of firm value
- Recovery in default is VT, not D. The gap is the sold put paying out
- Subordinating the debt does not change the argument — it only moves the strike

Source: CFA Institute, Curriculum Vol 7, Learning Module 9, §6 "Option Put–Call Parity Applications: Firm Value", Exhibit 9 and Equation 5, printed pp. 209–211.

SECTION 04

One-period binomial

Options need a model of the future because the payoff is kinked — the binomial tree is the simplest honest one

A forward payoff is linear, so it replicates without any view of the future. An option payoff bends at X, and that bend forces a model.

$$S_1^u = R_u \cdot S_0 \qquad S_1^d = R\_d \cdot S_0 \qquad (R_u > 1 > R\_d)$$

where: R_u and R_d are gross returns. q is the real-world probability of an up move — it is never needed. The SPREAD R_u − R_d is what carries the volatility assumption.

TWO OUTCOMES, NO MORE

Over one period the underlying goes to S_1^u or S_1^d. No third state. The curriculum calls it a Bernoulli trial. Everything else on the next four slides follows from arbitrage alone.

SPREAD = VOLATILITY

R_u − R_d is the model's entire view of volatility. Widen it — raise R_u, lower R_d, or both — and the range of possible payoffs widens, so BOTH the call and the put get more valuable.

ONE STEP → A TREE

Chain N one-period steps and you have an N-period tree with N + 1 terminal nodes. The curriculum stops at one period at Level I, but the machinery is identical at every node.

Source: CFA Institute, Curriculum Vol 7, Learning Module 10, §2 "Binomial Valuation" and §3 "The Binomial Model", Equations 1–2 and Exhibit 1, printed pp. 221–223.

Buy h shares against one short call and the portfolio stops caring which branch the tree takes

The hedge ratio is the binomial-world delta: the exact share count that makes a long-stock / short-call portfolio worth the same in both states.

$$h^* = (c_1^u - c_1^d) / (S_1^u - S_1^d)$$
$$V_1 = h^* S_1^u - c_1^u = h^* S_1^d - c_1^d$$

where: Choose h so the portfolio is worth the same either way. It is then riskless, so it must earn r — which gives $c_0 = h^*S_0 - V_1(1 + r)^{-1}$.*

THE FOUR-STEP RECIPE

1. Tree the underlying: $S_1^u = R_u S_0$ and $S_1^d = R\_dS_0$.
2. Tree the option: $c_1^u = \text{Max}(0, S_1^u - X)$ and $c_1^d = \text{Max}(0, S_1^d - X)$. For a put, $\text{Max}(0, X - S_1)$.
3. Compute h* from the payoff ratio, then $V_1 = h^*S_1^u - c_1^u$ (or the down branch — equal by construction).
4. V_1 is certain, so discount it at r and solve $c_0 = h^*S_0 - V_1(1 + r)^{-1}$.

➢ No probability appears anywhere. Only R_u, R_d, r and the option payoff.
➢ For a PUT, h* comes out negative — the curriculum notes that you then buy the option AND buy the underlying, rather than one against the other.

Source: CFA Institute, Curriculum Vol 7, Learning Module 10, §4 "Pricing a European Call Option", Equations 3–7 and Exhibits 2–3, printed pp. 223–226.

CFA LM 10: S_0 = 80, X = 100, up to 110 or down to 60, r = 5% — the delta hedge prices the call at EUR 4.57

Deliberately chosen numbers: the call is out of the money today ($S_0 < X$) but in the money in the up state. Every step of the recipe fires.

ONE-YEAR EUROPEAN CALL · THE DELTA-HEDGE ROUTE

S_0 = EUR 80, X = EUR 100, S_1^u = EUR 110 (R_u = 1.375), S_1^d = EUR 60 (R_d = 0.75), r = 5%, T = 1.

1. Option payoffs: c_1^u = Max(0, 110 − 100) = EUR 10; c_1^d = Max(0, 60 − 100) = 0
2. Hedge ratio: h* = (10 − 0) / (110 − 60) = 10 / 50 = 0.20 shares per call sold
3. Portfolio at T: V_1^u = 0.20 × 110 − 10 = 12; V_1^d = 0.20 × 60 − 0 = 12 ✓equal by construction
4. That 12 is certain, so discount it at r: V_0 = 12 / 1.05 = EUR 11.43
5. $c_0 = h^*S_0 - V_0$ = 0.20 × 80 − 11.43 = 16.00 − 11.43 = EUR 4.57

c_0 = EUR 4.57. Check: the 11.43 invested grows to 12 — exactly 5%, the risk-free rate.

Notice what never appears: no expected return on the stock, no risk premium, no view. The price came out of a hedge.

Source: CFA Institute, Curriculum Vol 7, Learning Module 10, §4, Exhibits 2–3 and Equation 7, printed pp. 223–226.

The same 4.57 falls out of a risk-neutral expectation — π is a pricing weight, not a forecast

π is built out of R_u, R_d and r alone. Nobody's opinion about the stock enters it, which is exactly why everyone can agree on it.

$$\pi = (1 + r - R_d) / (R_u - R_d)$$
$$c_0 = [\, \pi \cdot c_1^u + (1 - \pi) \cdot c_1^d \,] / (1 + r)^T$$

where: π is the risk-neutral probability of an up move; $1 - \pi$ of a down move. It sits strictly between 0 and 1 whenever $R_d - 1 < r < R_u - 1$, which is the no-arbitrage condition itself.

THE SAME TREE — THE RISK-NEUTRAL ROUTE

Reuse the canonical example: S_0 = EUR 80, X = EUR 100, $R_u = 1.375$, $R_d = 0.75$, r = 5%, c_1^u = EUR 10, $c_1^d = 0$.

1. $\pi = (1 + 0.05 - 0.75) / (1.375 - 0.75) = 0.30 / 0.625 = 0.48$, so $1 - \pi = 0.52$
2. Risk-neutral expected payoff = 0.48 × 10 + 0.52 × 0 = EUR 4.80
3. Discount one period at r: c_0 = 4.80 / 1.05 = EUR 4.57

Same 4.57. Faster route, and the only one that scales to an N-period tree.

Source: CFA Institute, Curriculum Vol 7, Learning Module 10, §5 "Risk Neutrality", Equations 9–10, printed pp. 230–231.

Real-world probabilities do not move the price; the width of the tree and the risk-free rate do

One-year options on a non-dividend stock: S_0 = USD 50, X = USD 55, r = 5%. Puts follow from parity. Only the first two rows differ.

Scenario	S_1^u	S_1^d	h*	π	Call c_0	Put p_0
±20% move — base case	60	40	0.25	0.625	2.98	5.36
±40% move — tree twice as wide	70	30	0.375	0.5625	8.04	10.42
±20%, bullish view q = 0.80	60	40	0.25	0.625	2.98	5.36
±20%, bearish view q = 0.20	60	40	0.25	0.625	2.98	5.36

READING THE TABLE

Rows 1 → 2: doubling the width of the tree lifts the call 2.7x, from 2.98 to 8.04, and the put from 5.36 to 10.42. Volatility is the factor that pays.

Rows 1, 3, 4: changing the real-world probability q from 0.50 to 0.80 to 0.20 leaves every price untouched — because q appears nowhere in either pricing route. The curriculum is blunt about it: a more optimistic outlook does not change the option price, because the seller can hedge the risk away with h* units of stock.

Puts here come straight from parity: $p_0 = c_0 - S_0 + X(1 + r)^{-T}$, and the risk-neutral route confirms them.

Source: CFA Institute, Curriculum Vol 7, Learning Module 10, §4 Example 1 "Hightest Capital", printed pp. 226–228, and §5 Example 2 "Hightest Capital (revisited)", printed pp. 231–232.

Push the steps to infinity and the tree becomes Black-Scholes — context beyond Vol 7, examined at Level II

Set $R_u = e^{\sigma\sqrt{\Delta t}}$ and $R_d = e^{-\sigma\sqrt{\Delta t}}$ at every step, let $\Delta t \to 0$, and the N-period tree converges to the closed-form price.

$$c_0 = S_0 \cdot N(d_1) - X \cdot e^{-rT} \cdot N(d_2)$$
$$d_{1,2} = \left[\ln(S_0/X) + (r \pm \sigma^2/2)T \right] / (\sigma\sqrt{T})$$

where: $N(\cdot)$ is the standard normal CDF; σ is the annualised volatility of continuously compounded returns. $N(d_2)$ is the continuous-time cousin of π.

WHY THE TREE COMES FIRST

A tree is pedagogy: every step is a visible replication and arbitrage argument. Black-Scholes is the same argument taken to a continuous limit — much harder to see, much cheaper to compute. Level I stops where the argument is still visible.

And the direction matters: the tree is not an approximation to Black-Scholes. Black-Scholes is a limit of the tree.

WHERE TREES STILL WIN

American and Bermudan exercise, path-dependent payoffs, discrete dividend dates — none of these fit a closed form. Plain European options get Black-Scholes; almost everything else on a real book gets a tree or a Monte Carlo.

Which is why the model you have just learned is not the toy version of the real one. It IS the real one, at N = 1.

[1] Black-Scholes does not appear in CFA Vol 7. Level I examines only the one-period binomial model.
[2] Trees remain standard for American exercise and discrete dividends; Black-Scholes admits neither.

Source: The formula and the up/down factors are NOT in CFA Vol 7 — shown as context. Curriculum note extended here: Learning Module 10, §5, printed p. 231.

KEY TAKEAWAYS

Deck 18 — Derivatives II: Options — Payoffs, Strategies & Pricing

- Payoffs at expiry: $c_T = \text{Max}(0, S_T - X)$, $p_T = \text{Max}(0, X - S_T)$. Profit subtracts the premium. Only the short call can lose an unlimited amount.
- Price = exercise value + time value. Exercise value discounts X to PV(X). Time value is positive and decays to zero at T.
- Moneyness is the holder's view: a call is ITM when $S > X$, a put when $S < X$. Bounds: $\text{Max}(0, S - PV(X)) < c \le S$ and $\text{Max}(0, PV(X) - S) < p \le X$.
- Six factors. Volatility and time to expiration move calls and puts the SAME way; S, X, r and income/carry move them opposite ways.
- Put-call parity: $S_0 + p_0 = c_0 + X(1+r)^{-T}$. It is a no-arbitrage identity, so it prices any one of the four positions from the other three.
- Synthetics: long stock = call − put + PV(X); covered call = long bond − put; equity in a levered firm = a call on the assets.
- Binomial: $h^* = (c_1^u - c_1^d)/(S_1^u - S_1^d)$ builds a riskless portfolio; its discounted cost pins c_0. h^* is negative for a put.
- Risk-neutral $\pi = (1 + r - R_d)/(R_u - R_d)$ gives the same price in one step. The real probability q never appears. Canonical answer: EUR 4.57.